

SIMATS ENGINEERING

ASSESSMENT TOOL 2

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COURSE NAME: Computer Architecture

COURSE CODE: CSA1202

QUESTIONS:5

Total Marks:25

ANNEXURE A

CONCEPT MAPPING

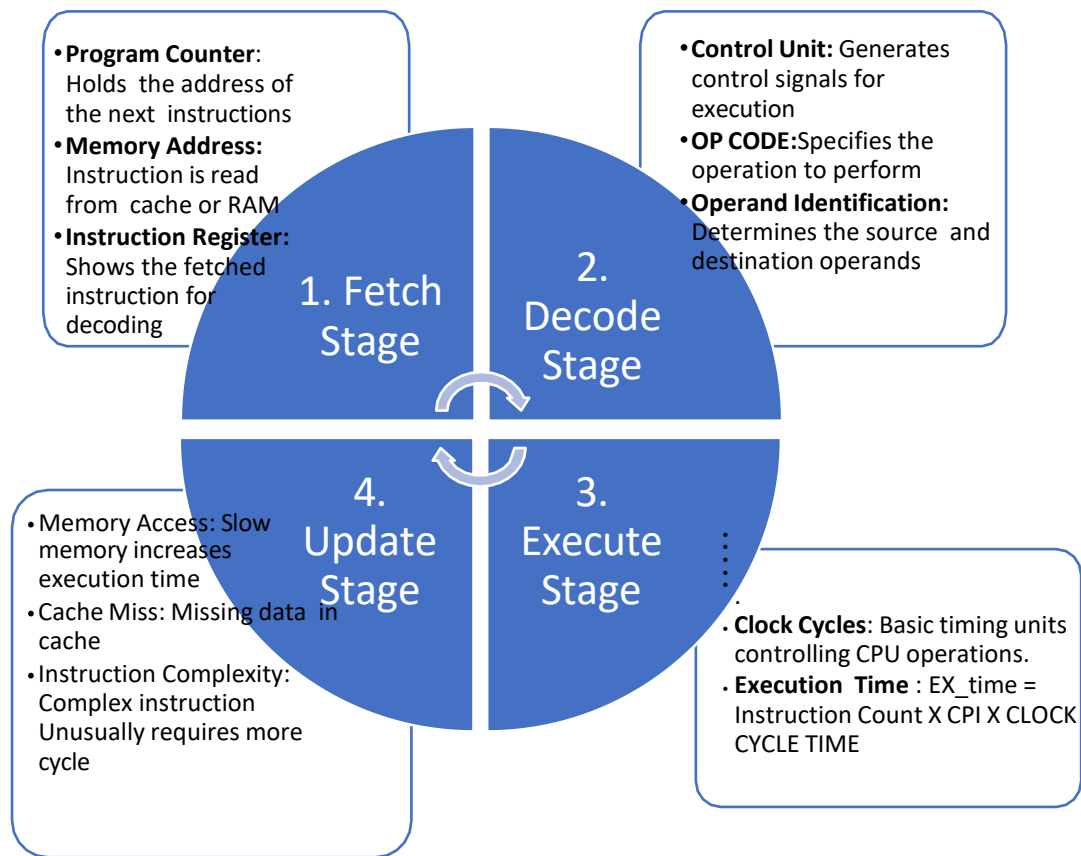
Code of Conduct:

I Lochana Sri R.S./ 192521348 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Lochana Sri R.S.

Signature of the student

1. Develop a comprehensive concept map linking the instruction execution cycle (fetch, decode, execute) with CPU performance parameters such as CPI, clock cycles, and execution time. Show how each stage contributes to overall execution time, and include connections to factors like memory access, instruction complexity, and pipeline stalls. Also, represent how benchmarking is used to evaluate performance.



2. Create a concept map comparing 8085 and 8086 architectures, including their instruction sets, register organization, addressing modes, and memory segmentation (in 8086). Clearly show how these architectural features influence program execution, multitasking capability, and efficiency in real-time applications.

8085 - MICROPROCESSOR:

An 8-bit microprocessor designed for simple embedded and control applications

Data bus : 8-bit , Address bus: 16-bit

Memory: Can access upto 64 KB

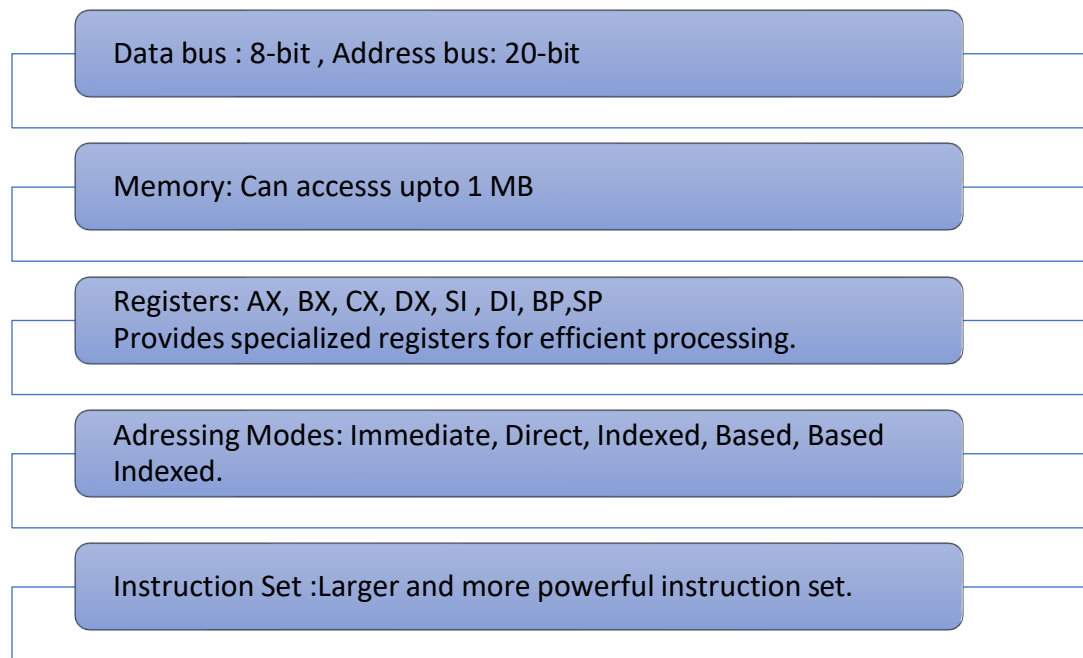
Registers: A, B, C, D, E, H, L
uses general purposes and accumulator registers.

Addressing Modes: Immediate, Direct, Register, Register Indirect

Instruction Set : Smaller Instruction set with fewer operations

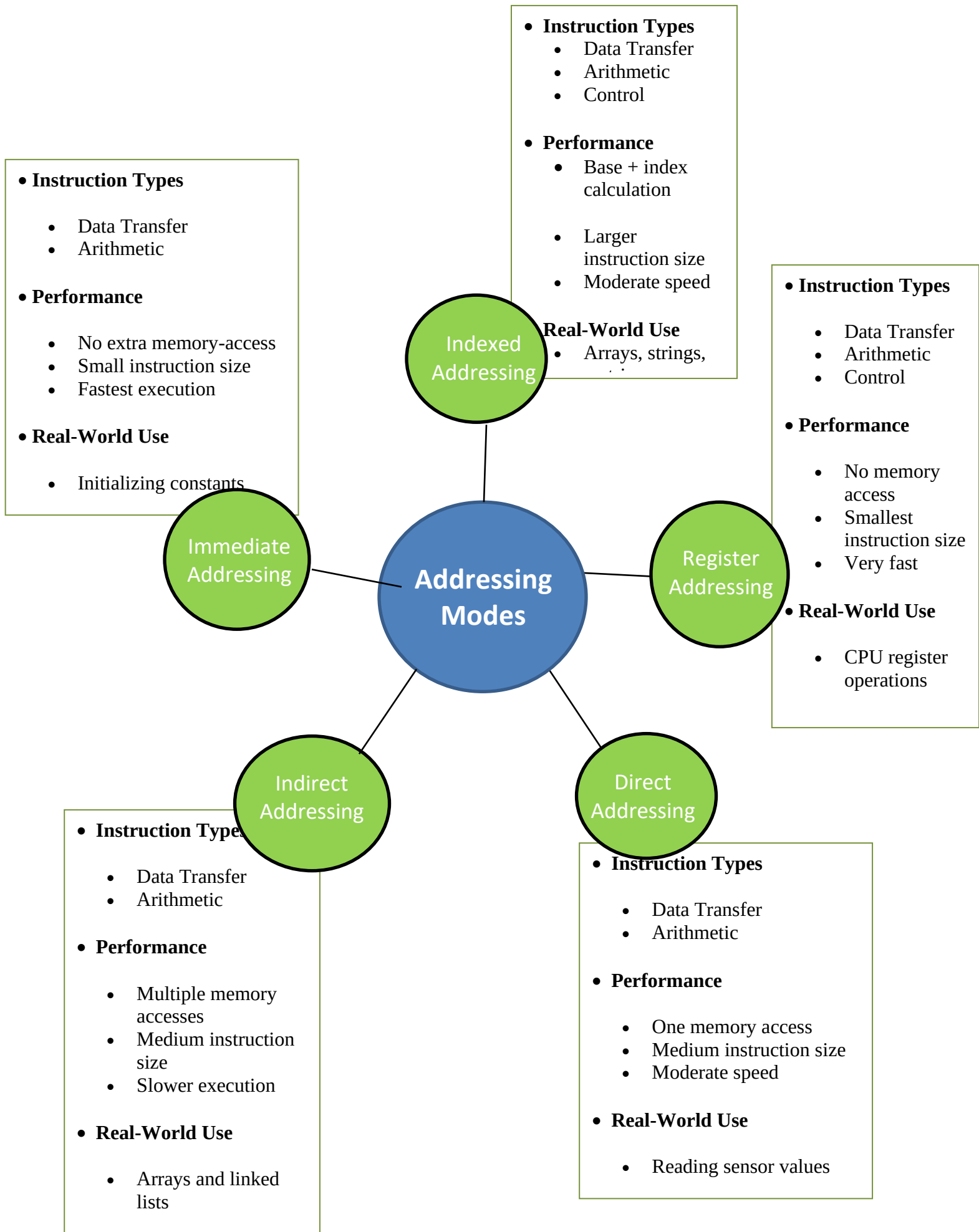
8086 - MICROPROCESSOR:

A 16-bit processor with advanced architecture for faster execution

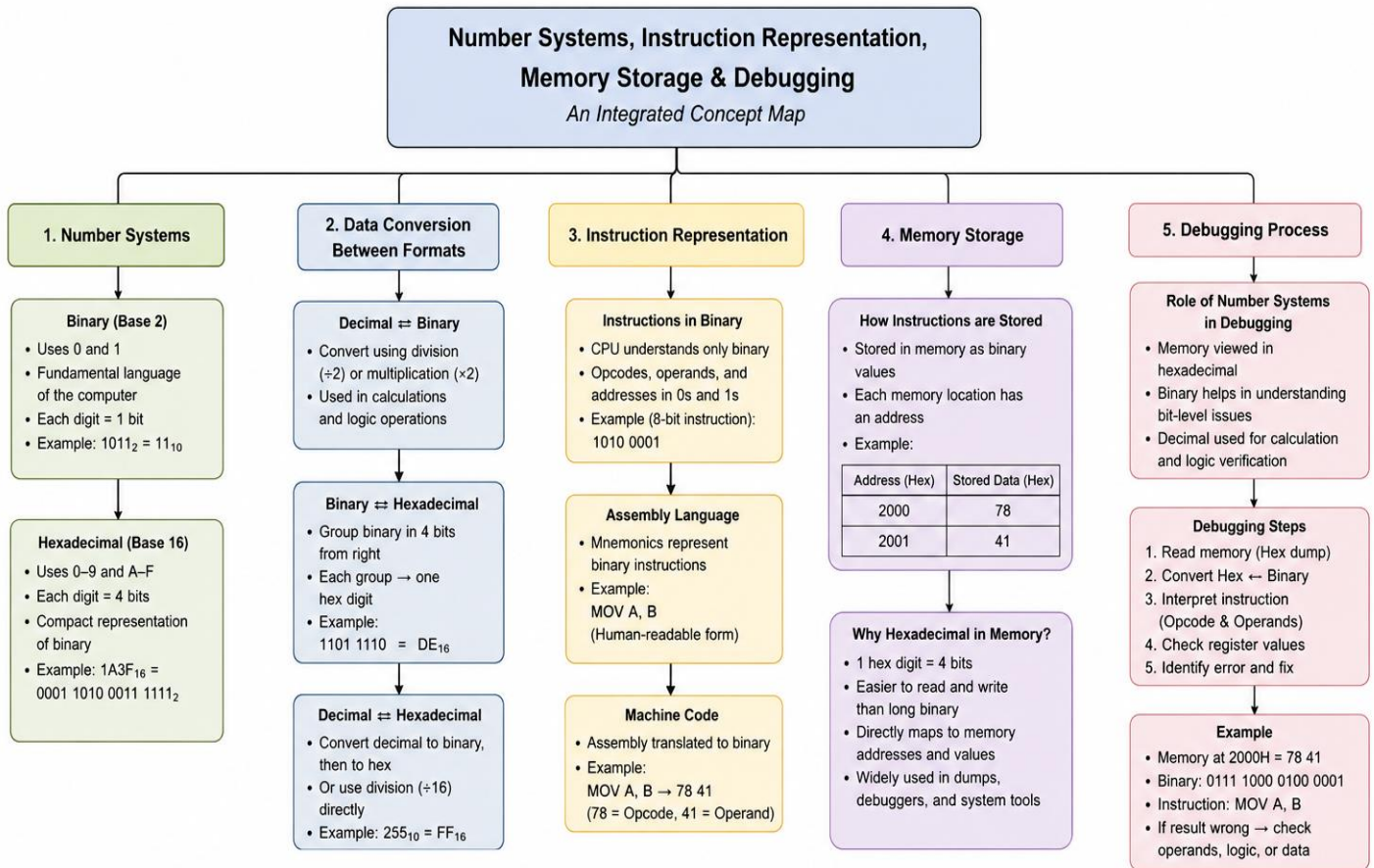


3. Draw a detailed concept map that connects various addressing modes (immediate, direct, register, indirect, indexed) with instruction types (data transfer, arithmetic, control instructions). Illustrate how each addressing mode affects memory access time, instruction size, and execution speed, and show real-world scenarios where specific addressing modes are preferred.

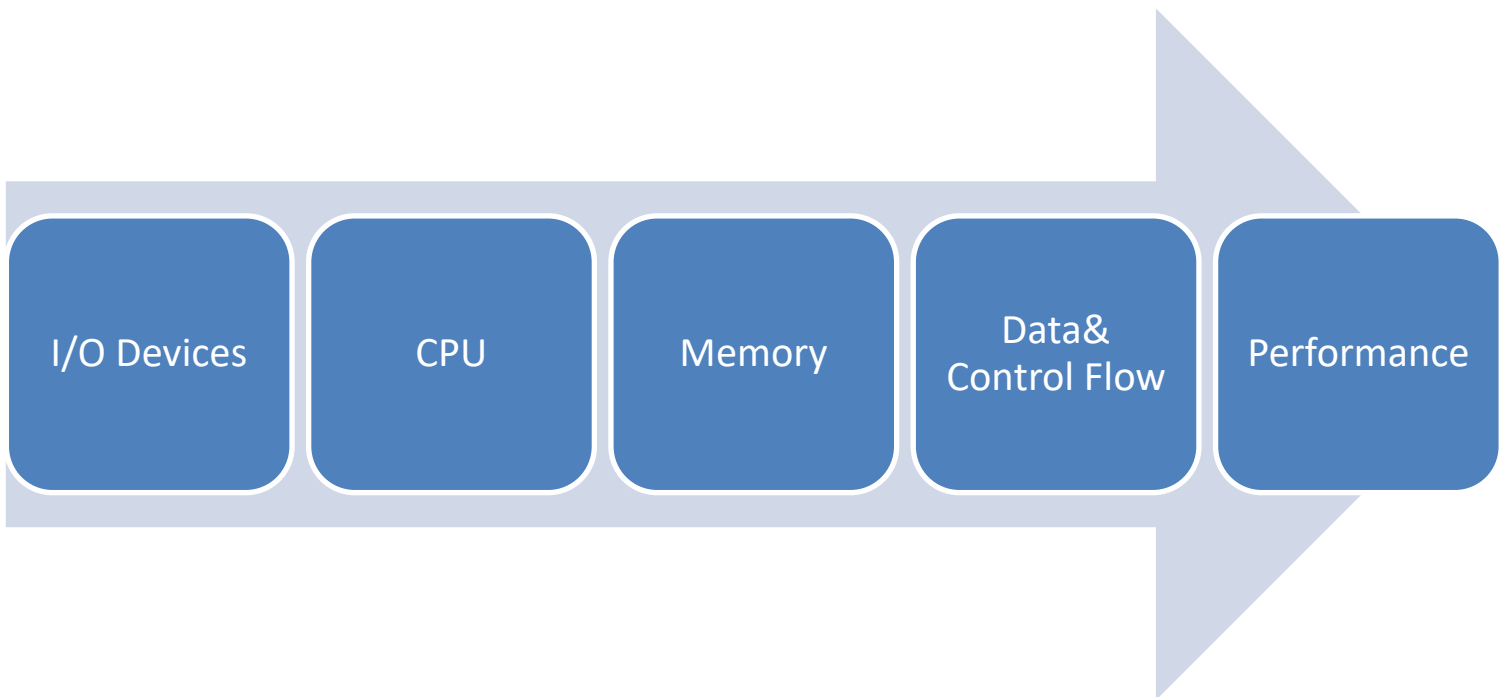
EXPLANATION: Addressing modes define how a processor accesses operands required for executing instructions. Different addressing modes affect **memory access time, instruction size, and execution speed**, making them suitable for different applications. Immediate and register addressing are faster because they reduce or eliminate memory access, while direct, indirect, and indexed addressing provide greater flexibility for accessing memory.



4. Prepare an elaborate concept map integrating number systems (binary and hexadecimal) with instruction representation, memory storage, and debugging processes. Show how data is converted between formats, how instructions are stored and interpreted by the CPU, and why hexadecimal representation is commonly used in low-level programming and system design.



5. Construct a detailed concept map showing the relationship between CPU (ALU, Control Unit, Registers), memory (cache, RAM), and I/O devices. Extend the map to include single-bus, multi-bus, and hierarchical bus structures, and clearly illustrate how data and control signals flow among these components during execution. Indicate possible points of delay or bottlenecks and how different bus structures help in improving performance.



I/O Devices

- Keyboard
- Mouse
- Monitor
- Printer

CPU

ALU

- Arithmetic Operations
- Logical Operations

Control Unit

- Controls Execution
- Generates Control Signals

Registers

- Temporary Data Storage

- Fast Access

Memory

Cache

- Very Fast
- Frequently Used Data

RAM

- Main Memory
- Stores Programs & Data

Bus Structures

Single Bus

- One communication path
- Low cost
- More congestion

Multi Bus

- Multiple communication paths
- Faster transfer
- Less waiting

Hierarchical Bus

- Local & System buses
- Better scalability
- High performance

Data Flow

I/O Devices



CPU Fetches Data



Memory Supplies Instructions



Bus Transfers Data & Control Signals



CPU Executes



Result Stored in Memory



Output Device Displays Result

Delay / Bottlenecks

- Cache Miss
- Slow RAM Access
- Bus Contention
- I/O Waiting Time

Performance Improvement

Single Bus

→ Simple Design

Multi Bus

→ Parallel Transfers

Hierarchical Bus

→ Reduced Traffic

→ Faster Communication